



9c. Generative Artificial Intelligence

Never Stand Still

Faculty of Engineering

COMP9444 9c

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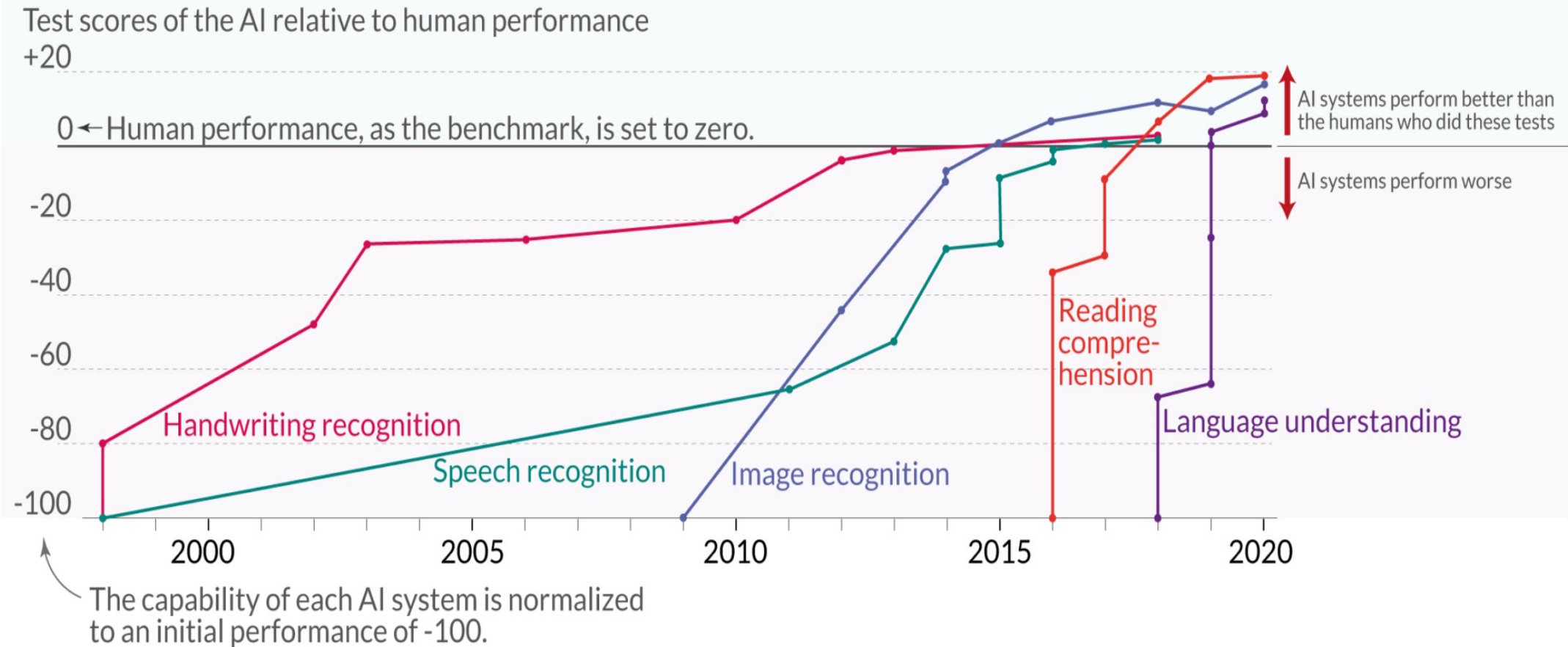
Do not remove this notice

Key-ideas covered throughout the course

- Neural Networks and their training
 - Perceptron and Perceptron Learning Algorithm
 - Multi-Layer Neural Networks and Backpropagation Algorithm
- Convolutional Neural Networks (CNNs)
 - Building blocks of a typical CNN architecture
 - LeNet, AlexNet, VGG-Net, ResNet, GoogLeNet, DenseNet, ...
- Recurrent Neural Networks (RNNs)
 - Vanilla RNNs
 - Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU)
 - Transformer model
 - Bidirectional Encoder Representations from Transformers (BERT)
- Reinforcement Learning
 - Value function learning (TD-Learning, Q-Learning)
 - Policy learning
 - Actor-Critic
- Autoencoders, Variational Autoencoder (VAE)
- Generative Adversarial Networks (GANs)

The AI context

Language and image recognition capabilities of AI systems have improved rapidly



Data source: Kiela et al. (2021) – Dynabench: Rethinking Benchmarking in NLP
OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the author Max Roser

What is Generative Artificial Intelligence?

- Set of algorithms that can generate:
 - Images (GANs)
 - Text (Chat-GPT)
 - Audio (OpenAI Whisper)
 - Videos (Synthesia)
 - Code (GitHub CoPilot)
 - Simulations (CARLA – Open-source simulator for autonomous driving research)
- Guided by:
 - Training sample
 - User specifications
- Has huge potential to drastically change the way we approach content creation.

Generative AI tools

➤ Chat-GPT



What is Artificial General Intelligence



Artificial General Intelligence (AGI) refers to a type of artificial intelligence that possesses the ability to understand, learn, and apply knowledge in a way that is similar to human intelligence. Unlike narrow or specialized AI, which is designed to perform specific tasks or solve particular problems, AGI aims to exhibit general intelligence across a wide range of tasks and domains.



Characteristics of Artificial General Intelligence:

1. **Flexibility:** AGI can adapt and apply its knowledge to various tasks and situations, even those it has not encountered before. It can learn from experiences and use that knowledge to solve new problems.
2. **Reasoning and Understanding:** AGI can reason, understand, and make sense of complex information. It can analyze and process data, draw logical conclusions, and understand cause-and-effect relationships.
3. **Learning:** AGI is capable of continuous learning and self-improvement. It can acquire new skills, knowledge, and capabilities through experience and interaction with its environment.
4. **Creativity:** AGI can demonstrate creative thinking, innovation, and the ability to generate novel and original ideas or solutions.
5. **Generalization:** AGI can apply its knowledge from one domain to another, allowing it to transfer skills and understanding.
6. **Communication:** AGI can effectively communicate with humans and other AI systems, both

🔄 Regenerate response

Send a message



Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts. [ChatGPT May 24 Version](#)

Generative AI tools

➤ Stable Diffusion

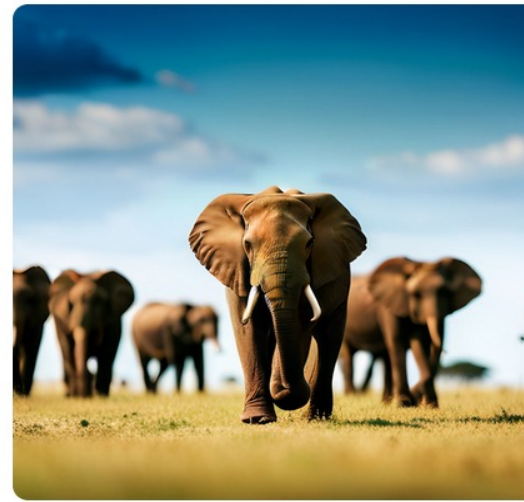
stability.ai

Stable Diffusion XL

Create and inspire using the worlds fastest growing open source AI platform.

With Stable Diffusion XL, you can create descriptive images with shorter prompts and generate words within images. The model is a significant advancement in image generation capabilities, offering enhanced image composition and face generation that results in stunning visuals and realistic aesthetics.

Stable Diffusion XL is currently in beta on DreamStudio and other leading imaging applications. Like all of Stability AI's foundation models, Stable Diffusion XL will be released as open source for optimal accessibility in the near future.



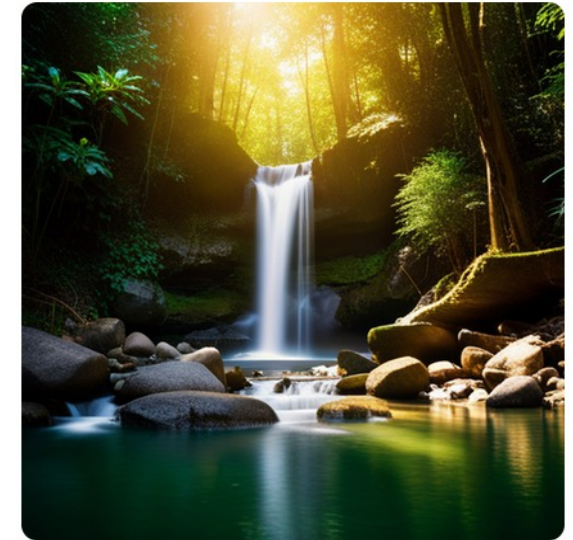
Prompt: Glimpses of a herd of wild elephants crossing a savanna



Prompt: Vintage hot rod with custom flame paint job



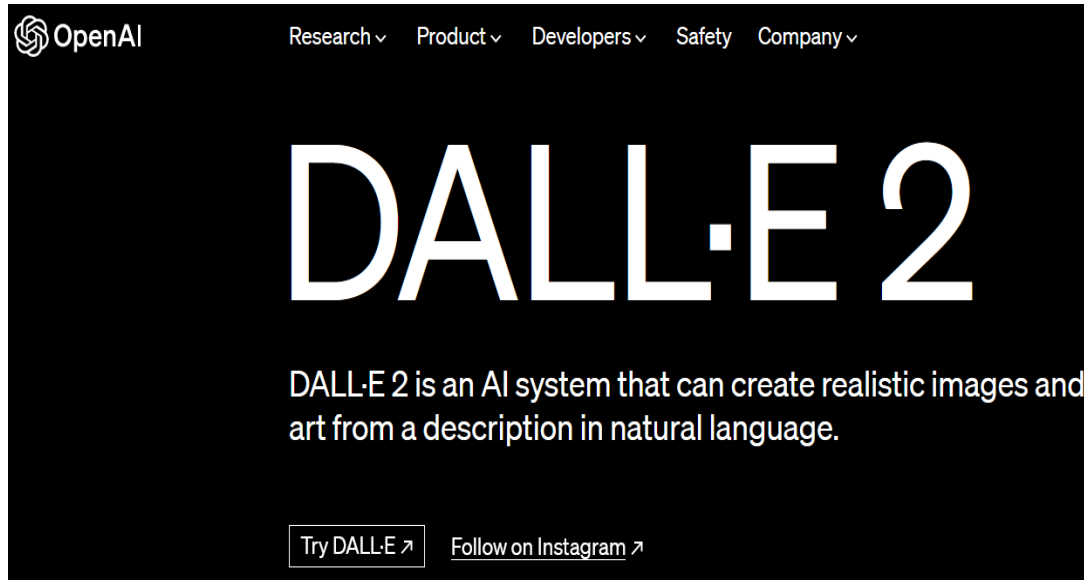
Prompt: Ancient, mysterious temple in a mountain range, surrounded by misty clouds and tall peaks



Prompt: Beautiful waterfall in a lush jungle, with sunlight shining through the trees

Generative AI tools

➤ DALL-E



Prompt: A hand-drawn sailboat circled by birds on the sea at sunrise



Prompt: A photograph of a sunflower with sunglasses on in the middle of the flower in a field on a bright sunny day



Prompt: A cartoon of a cat catching a mouse



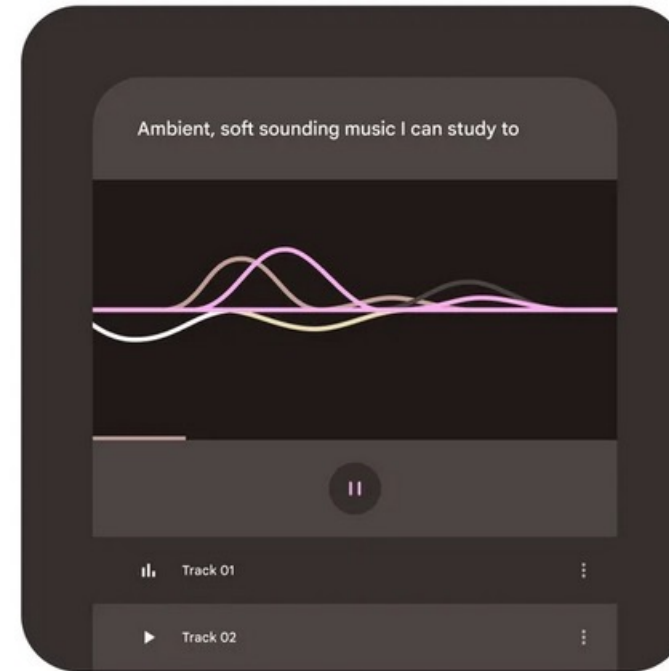
Prompt: 3D render of a cute tropical fish in an aquarium on a dark blue background, digital art

Generative AI tools

➤ MusicLM

MusicLM

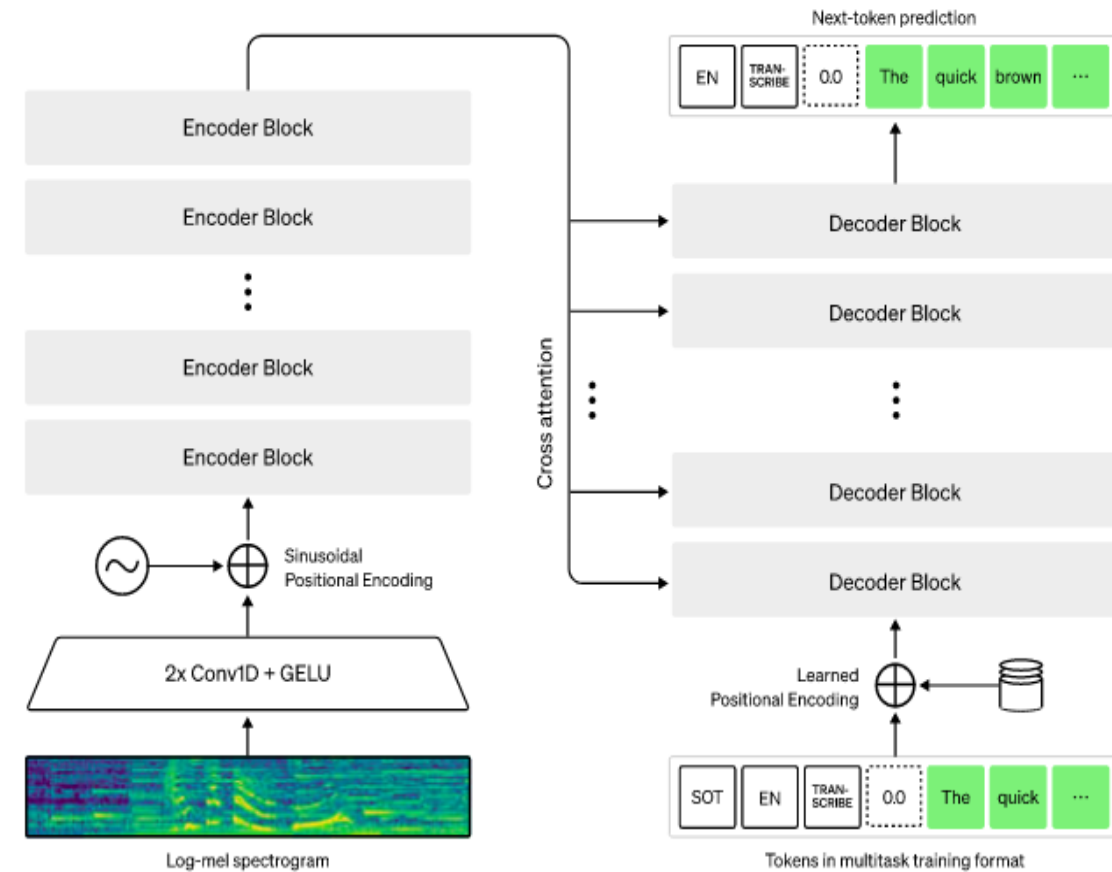
Describe a musical idea and hear it come to life with AI



Generative AI tools

➤ OpenAI Whisper

- An **Automatic Speech Recognition (ASR)** system trained on 680,000 hours of multilingual and multitask data.
- Simple **end-to-end Encoder-Decoder Transformer architecture**
- **Process:** Input audio is split into 30-second chunks, converted into a log-Mel spectrogram, and then passed into an encoder. A decoder is trained to predict the corresponding text caption, intermixed with special tokens that direct the single model to perform tasks such as language identification, phrase-level timestamps, multilingual speech transcription, and to-English speech translation.



Generative AI tools

➤ OpenAI Whisper

Whisper examples:

K-Pop ▾



While darkness was my everything
I ran so hard that I ran out of breath
Never say time's up
Like the end of the boundary
Because my end is not the end

Whisper examples:

Accent ▾



One of the most famous landmarks on the Borders, it's three hills and the myth is that Merlin, the magician, split one hill into three and left the two hills at the back of us which you can see. The weather's never good though, we stay on the Borders with the mists on the Yildens, we never get the good weather and as you can see today there's no sunshine, it's a typical Scottish Borders day.

Note: Whisper transcribed "Eildons" as "Yildens"

Whisper examples:

French ▾



Whisper is an automatic speech recognition system based on 680,000 hours of multilingual and multitasking data collected on the Internet. We establish that the use of such a number of data is such a diversity and the reason why our system is able to understand many accents, regardless of the background noise, to understand technical vocabulary and to successfully translate from various languages into English. We distribute as a free software the source code for our models and for the inference, so that it can serve as a starting point to build useful applications and to help progress research in speech processing.

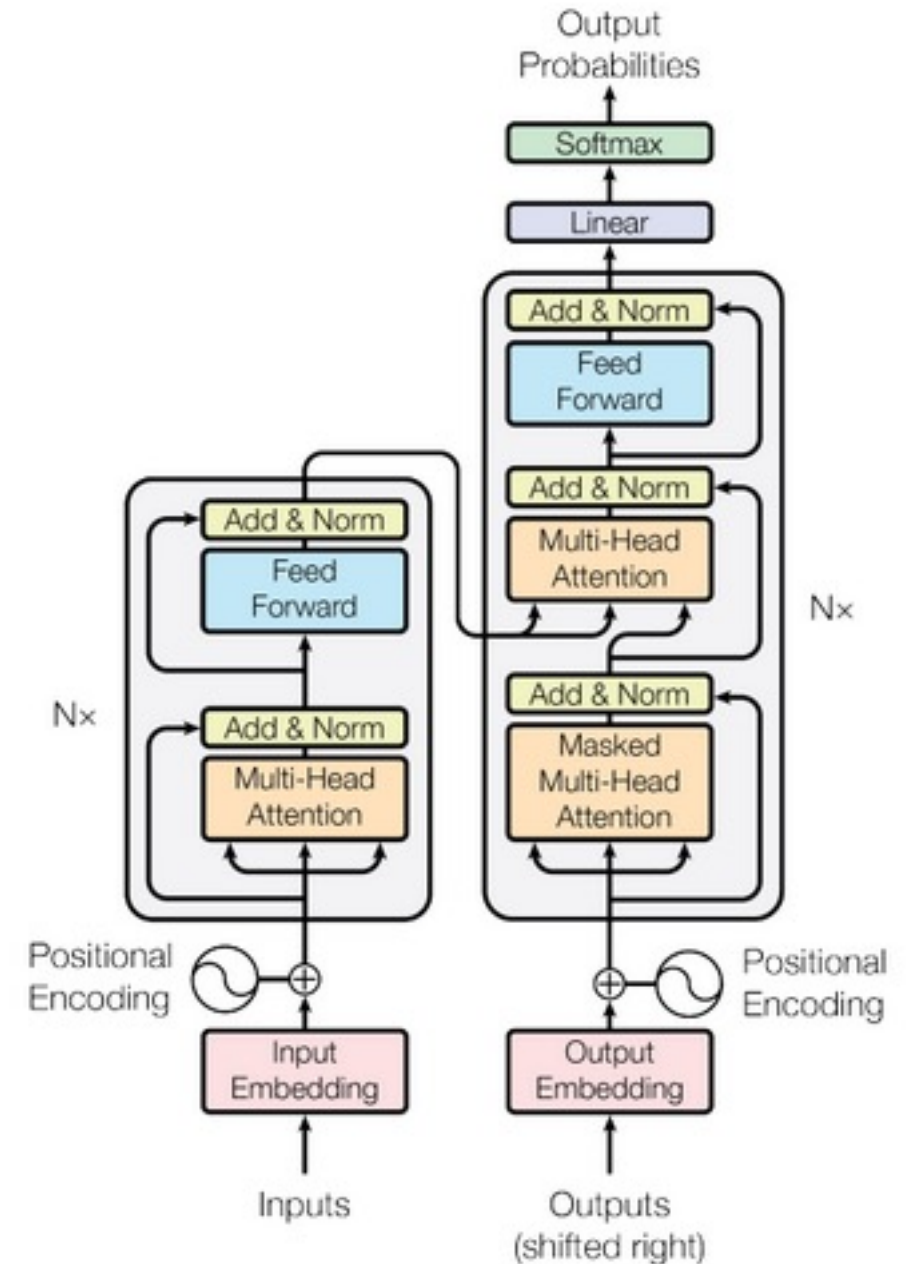
ChatGPT

- It is a powerful text-generating dialogue system that can generate humanlike responses to inputs from users.
- Based on Generative Pre-trained Transformer (GPT) architecture
- It is trained on vast data from the internet
- It can accomplish a variety of NLP tasks such as translation, answering questions, sentence completion, etc.
- Let's check GPT-4 capabilities
<https://openai.com/gpt-4>

Transformers

- Attention is All You Need
- Novel architecture relies entirely on **self-attention** to compute representations of its input and output without using sequential RNNs or convolutions.
- Aim is to solve seq2seq tasks while handling long-range dependencies

“Griezmann’s announcement comes as a bit of a shock. After enduring the drama surrounding his potential last summer, many thought he was committed to Atletico for more than a year, but the Frenchman seems to have changed his mind.”



GPT-3

- Key idea: Improve task-agnostic few-shot performance
- Evaluation on various NLP tasks under few-shot learning, one-shot learning, and zero-shot learning demonstrates GPT-3 promising results
- Technical details:
 - An autoregressive language model
 - GPT-3 has 96 layers with each layer having 96 attention heads
 - trained on datasets with 500 billion tokens
 - Word embedding size of 12888
 - Context window size is of 2048 tokens
 - Uses alternating dense and locally banded sparse attention patterns
- Compute:
 - Trained on more than 576 GB of text data including common crawl, Books, and Wikipedia
 - About 175 billion parameters
 - costs OpenAI around \$4.6 million

Language Models are Few-Shot Learners

Tom B. Brown*	Benjamin Mann*	Nick Ryder*	Melanie Subbiah*	
Jared Kaplan†	Prafulla Dhariwal	Arvind Neelakantan	Pranav Shyam	Girish Sastry
Amanda Askell	Sandhini Agarwal	Ariel Herbert-Voss	Gretchen Krueger	Tom Henighan
Rewon Child	Aditya Ramesh	Daniel M. Ziegler	Jeffrey Wu	Clemens Winter
Christopher Hesse	Mark Chen	Eric Sigler	Mateusz Litwin	Scott Gray
Benjamin Chess	Jack Clark	Christopher Berner		
Sam McCandlish	Alec Radford	Ilya Sutskever	Dario Amodei	
OpenAI				

GPT-3

- **Fine-Tuning:** Updating the weights of a pre-trained model by training on a supervised dataset specific to the desired task.
- **Few-Shot Learning:** Refers to the setting where the model is given a few demonstrations of the task at inference time as conditioning, but no weight updates are allowed.
- **One-Shot Learning:** In this setting, only one demonstration is allowed
- **Zero-Shot Learning:** In this setting, a model can learn to recognize things that it hasn't explicitly seen before during training. The idea behind is how humans can naturally find similarities between data classes, in the same way, training the machines to identify.

GPT-3

Zero-shot, one-shot and few-shot, contrasted with traditional fine-tuning

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



GPT-3

- Sizes, architectures, and learning hyper-parameters of GPT-3 models

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

- Datasets used to train GPT-3

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

GPT-3 Tasks

- **Language modelling:** Calculated zero-shot perplexity on the Penn Tree Bank (PTB) dataset
- **LAMBADA:** The LAMBADA dataset tests the modelling of long-range dependencies in text – the model is asked to predict the last word of sentences which require reading a paragraph of context.
- The **StoryCloze 2016 dataset**, which involves selecting the correct ending sentence for five-sentence long stories.
- The **HelloSwag dataset** involves picking the best ending to a story or set of instructions.
- **Closed Book Question Answering:** Measuring GPT-3's ability to answer questions about broad factual knowledge.
- **Translation:** Translation English, French, German, Romanian
- **Reading Comprehension**
- **SuperGLUE** (prominent evaluation framework for research towards language understanding)
- **Natural Language Inference (NLI)**
- **SAT Analogies**
- **Synthetic and Qualitative tasks** (addition, subtraction, ...)
- ... and more ...

GPT-3 Results

- Results on language modelling, Cloze, and completion Tasks

Setting	LAMBADA (acc)	LAMBADA (ppl)	StoryCloze (acc)	HellaSwag (acc)
SOTA	68.0 ^a	8.63 ^b	91.8^c	85.6^d
GPT-3 Zero-Shot	76.2	3.00	83.2	78.9
GPT-3 One-Shot	72.5	3.35	84.7	78.1
GPT-3 Few-Shot	86.4	1.92	87.7	79.3

- Results on three open-domain QA tasks

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP ⁺ 20]	44.5	45.5	68.0
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	68.0
GPT-3 Few-Shot	29.9	41.5	71.2

GPT-4

- Following the research path from GPT, GPT-2, and GPT-3, GPT-4 leverages more data and more computation to create increasingly sophisticated and capable language models.
- Still known limitations: social biases, hallucinations, and adversarial prompts

GPT-4 Technical Report

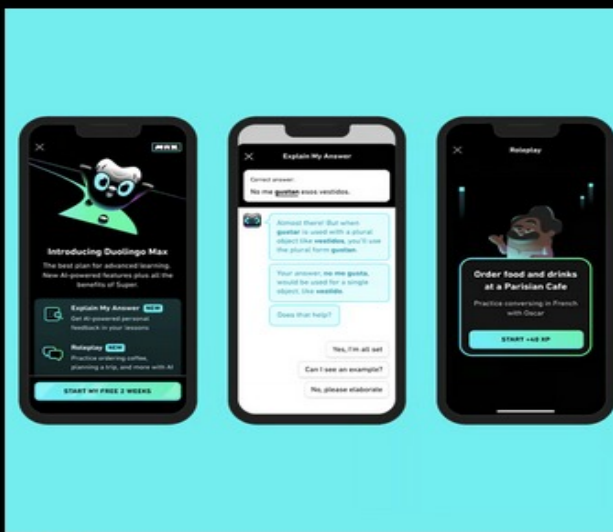
OpenAI*

Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

GPT-4

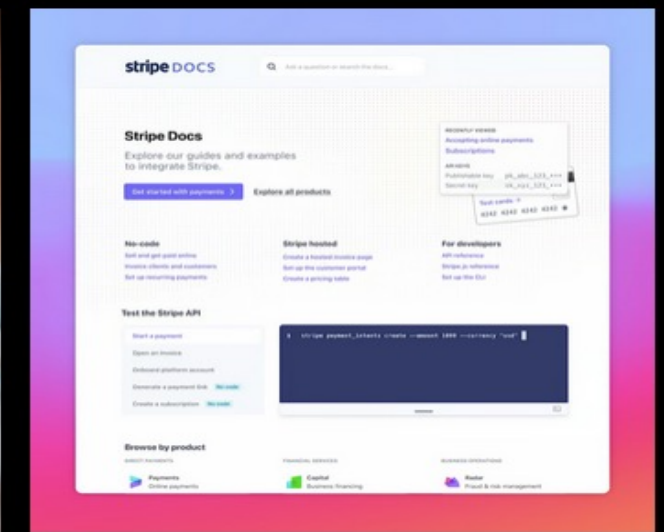
➤ Applications



Duolingo
GPT-4 deepens the conversation on Duolingo.



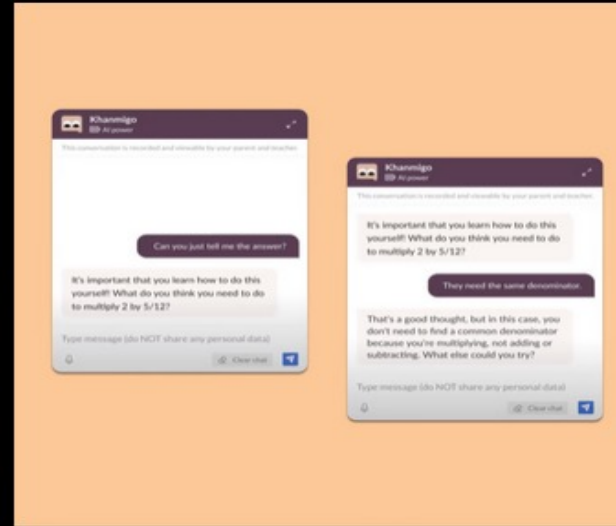
Be My Eyes
Be My Eyes uses GPT-4 to transform visual accessibility.



Stripe
Stripe leverages GPT-4 to streamline user experience and combat fraud.



Morgan Stanley
Morgan Stanley wealth management deploys GPT-4 to organize its vast knowledge base.



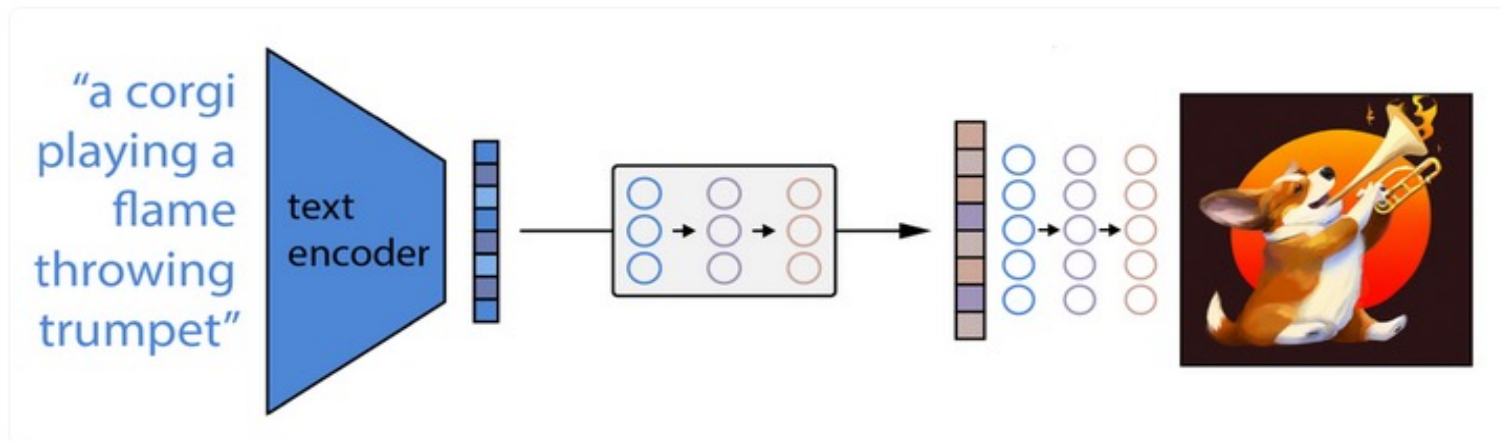
Khan Academy
Khan Academy explores the potential for GPT-4 in a limited pilot program.



Government of Iceland
How Iceland is using GPT-4 to preserve its language.

DALL-E 3

- Capabilities:
 - can generate images from text
 - can insert new features or styles in images to modify them
- Based on **CLIP** (Contrastive Language-Image Pre-training)



a propaganda poster depicting a cat dressed as french emperor napoleon holding a piece of cheese

DALL-E 3

- DALL-E 3 understands significantly more nuance and detail than DALL-E 2.



Vision Transformer (ViT)

- Transformers showed great performance on variety of NLP tasks.
- How to use Transformers (self-attention) for vision tasks?
- Transformer applied to image patches-
similar to NLP, but with patches instead
of words

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*},
Xiaohua Zhai^{*}, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer,
Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}

^{*}equal technical contribution, [†]equal advising

Google Research, Brain Team

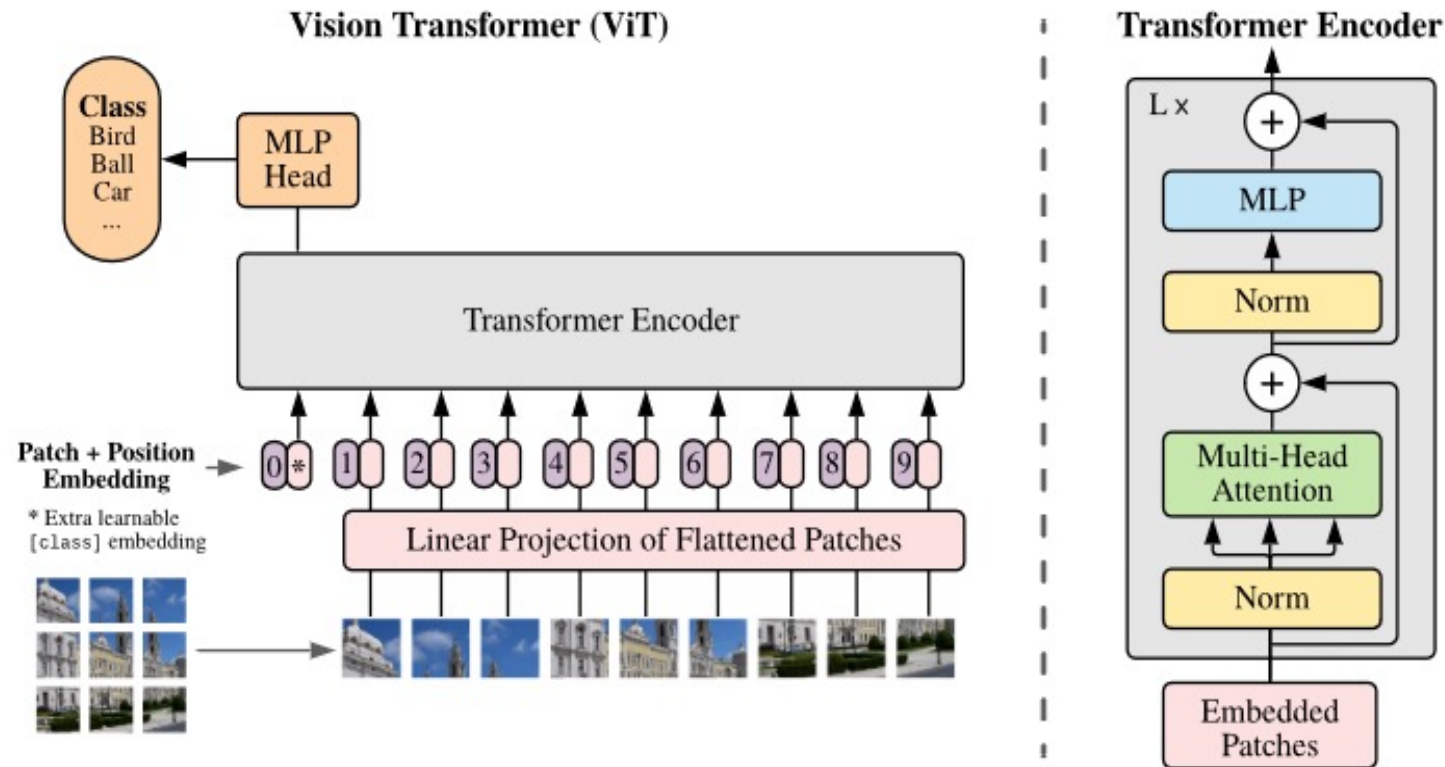
{adosovitskiy, neilhoulby}@google.com

ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train. 

Vision Transformer (ViT) method

- Split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder.



Vision Transformer (ViT) experimental setup

➤ Pre-training datasets

- ImageNet ILSVRC-2012: 1000 classes, 1.3 million images
- ImageNet-21k: 21000 classes, 14 million images
- JFT: 18000 classes, 303 million images

*Approximate number of classes and images

➤ Fine-tuning datasets

- ImageNet- both original validation labels and Reassessed Labels (ReaL)
- CIFAR-10/100: 60000 images each
- Oxford-IIIT Pets: 7400 images
- Oxford Flowers-102: 7100 images
- VTAB: natural, specialized, structured

➤ Pre-processing

- Pre-training: image cropped, random horizontal mirroring, resize to 224 x 224
- Fine-tuning: images resized to 448 x 448, random crop of 384 x 384
- Random horizontal flips

Vision Transformer (ViT)

➤ ViT variants

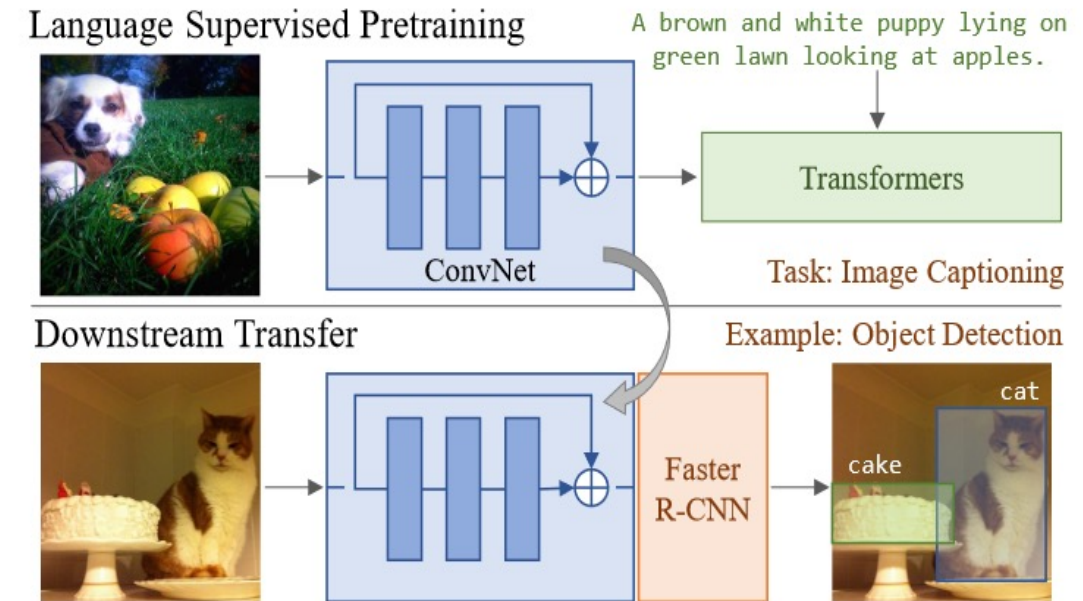
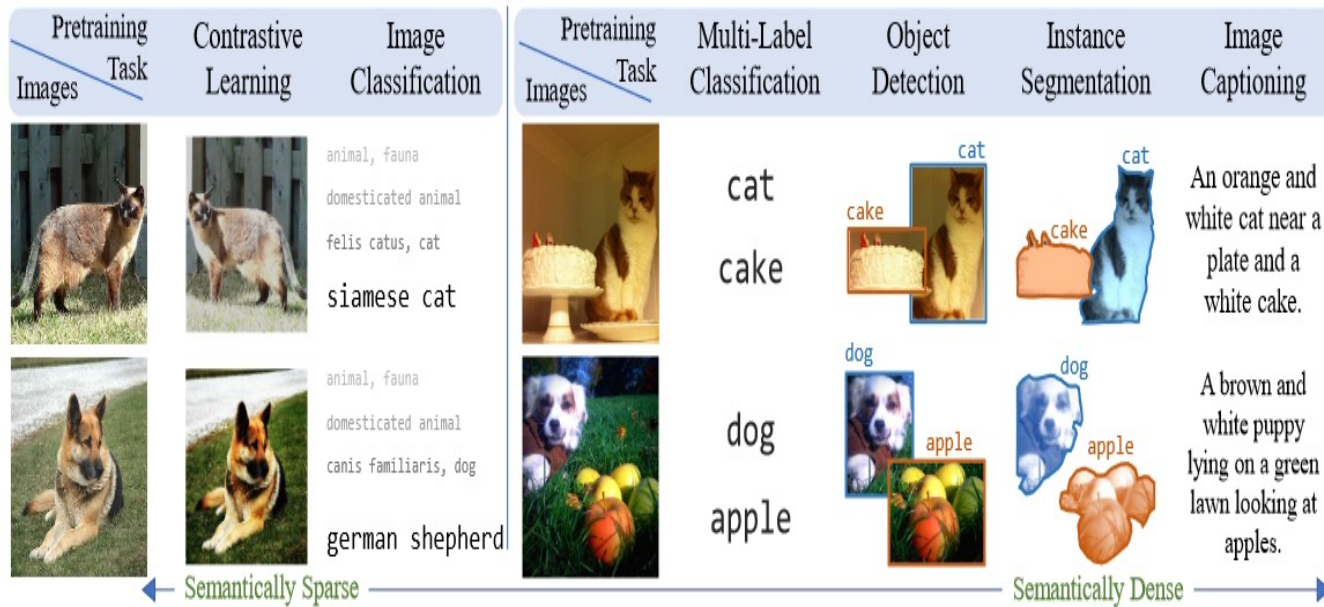
Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

➤ Results

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet Real	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

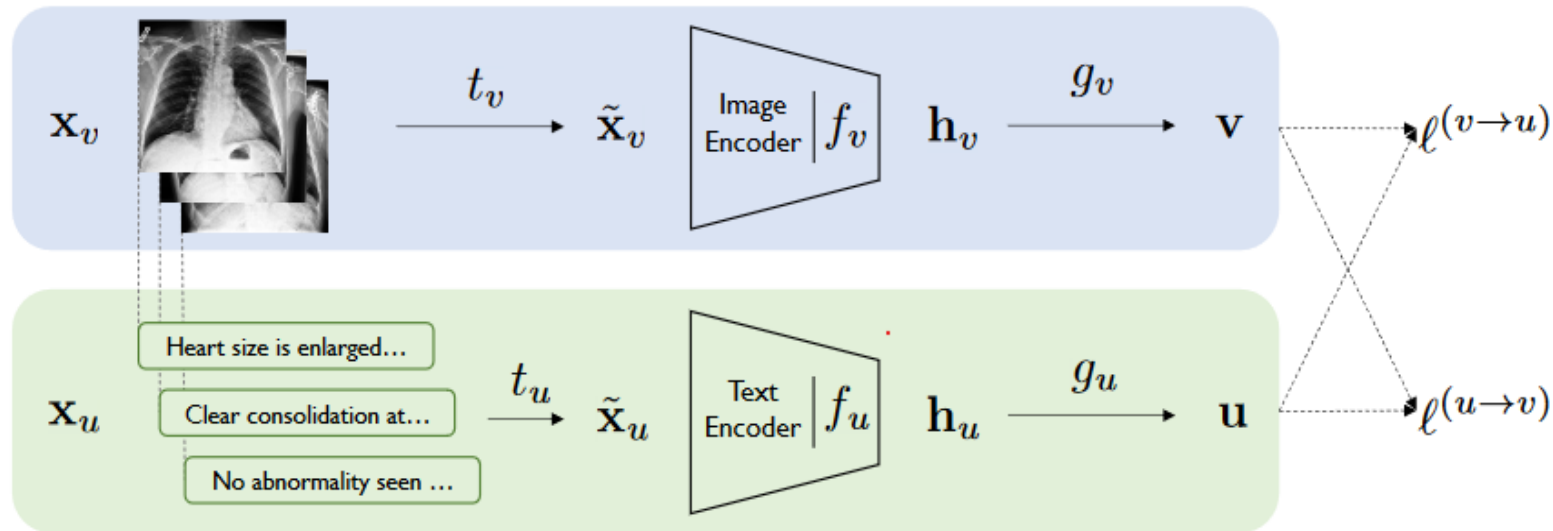
CLIP uses prior work of VirTex and ConVIRT

- VirTex: Learning Visual Representations from Textual Annotations
- Prevailing paradigm: Pretrain a CNN and then transfer the learned features to downstream tasks.
- Using textual features to learn visual features require fewer images than other approaches.



CLIP uses prior work of VirTex and ConVIRT

- ConVIRT: Contrastive Learning of Medical Visual Representations from paired images and text
- Proposes unsupervised strategy to learn visual representations by exploiting naturally occurring paired descriptive text (both in natural and medical domain)
- Maximizes the agreement between the true image-text representation pairs with bidirectional losses



CLIP (Contrastive Language-Image Pre-training)

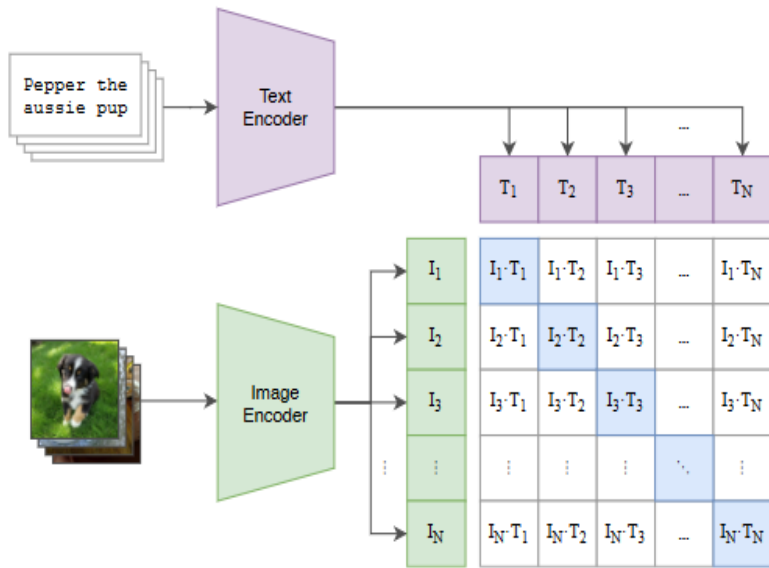
- CLIP learns visual concepts from natural language supervision
- Training process:
 1. All images and their associated captions are passed through respective encoders (image and text) to map images and text into n-dimensional vector
 2. Compute cosine similarity for each (image, text) pair
 3. During training, maximize the cosine similarity between correct encoded (image, text) pairs and minimize the cosine similarity between incorrect encoded (image, text) pairs.
- Let's visualize the training process

<https://www.assemblyai.com/blog/how-dall-e-2-actually-works/>

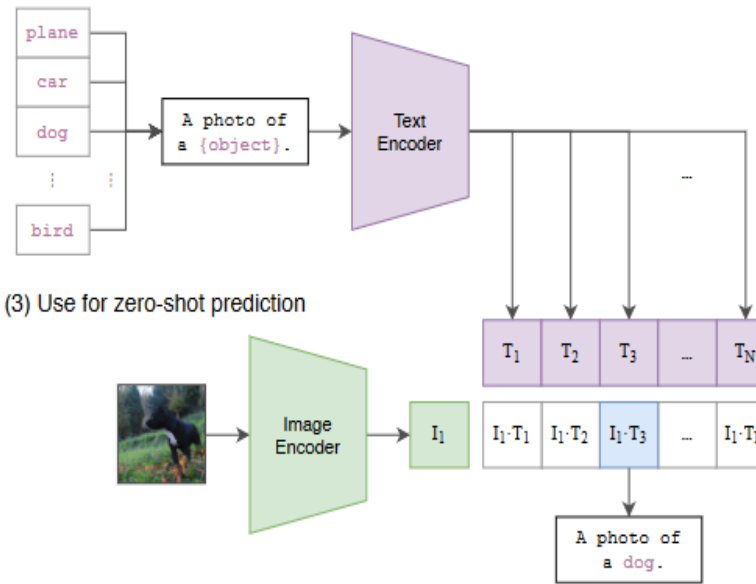
CLIP model

- CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples.
- At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Pseudocode of core implementation of CLIP

```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

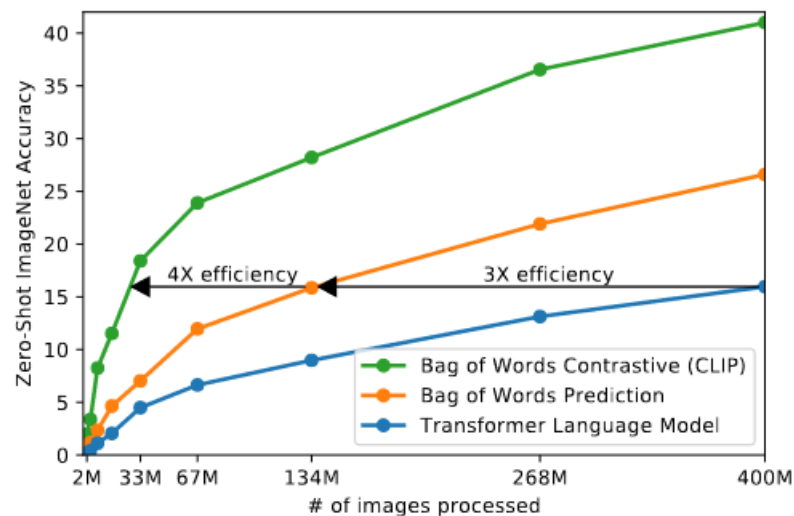
# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

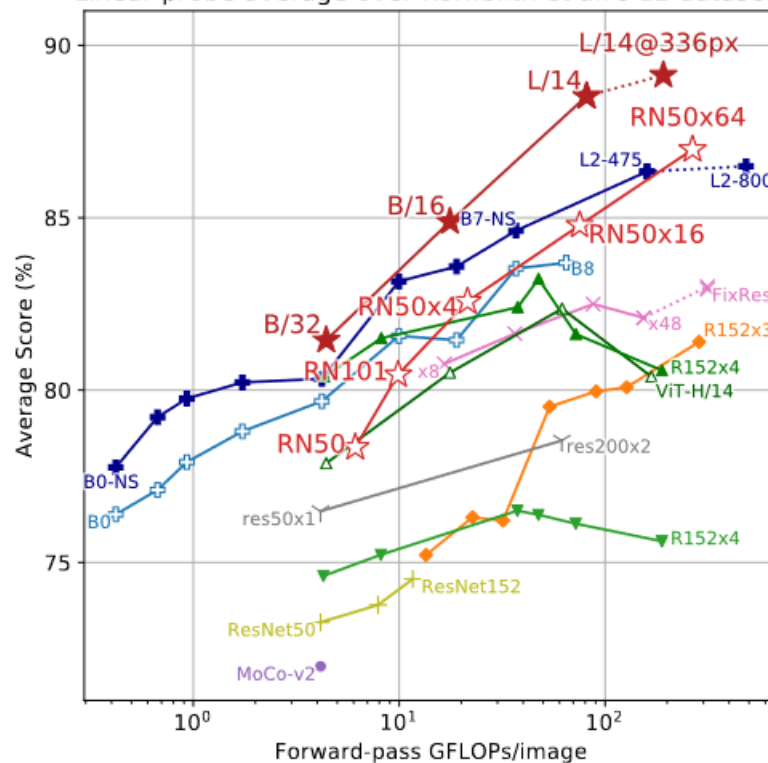
CLIP results

Linear probe performance of CLIP models in comparison to state-of-the-art computer vision models

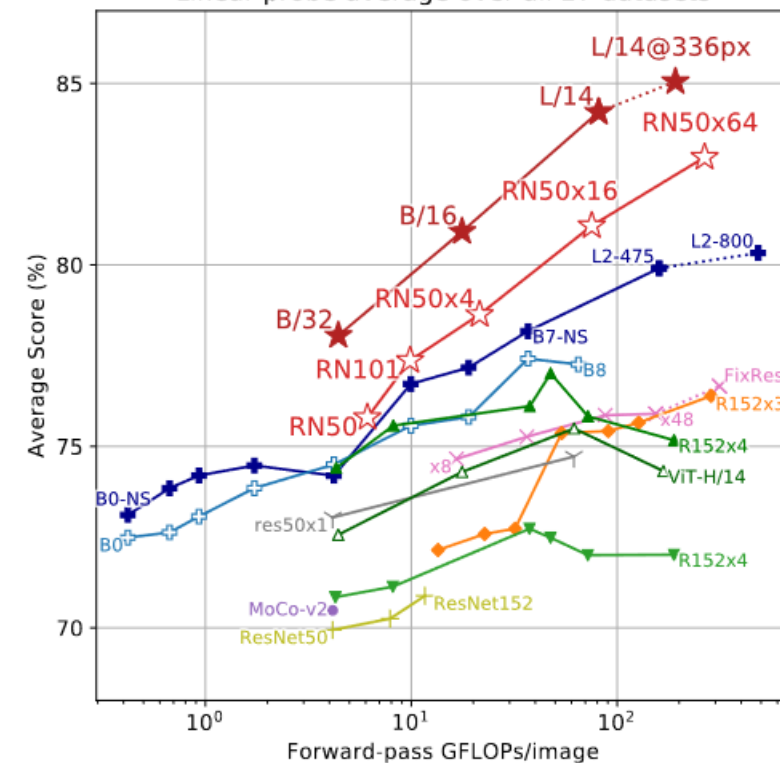
CLIP is much more efficient at zero-shot transfer than image caption baseline



Linear probe average over Kornblith et al.'s 12 datasets



Linear probe average over all 27 datasets



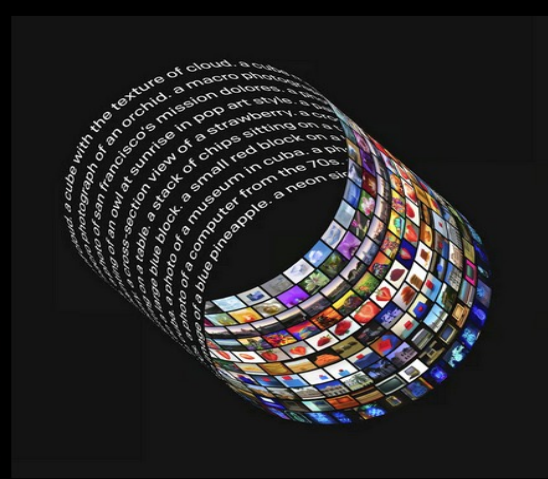
- CLIP-ViT
- CLIP-ResNet
- EfficientNet-NoiseStudent
- EfficientNet
- Instagram-pretrained
- SimCLRv2
- BYOL
- MoCo
- ViT (ImageNet-21k)
- BiT-M
- BiT-S
- ResNet

DALLE-3 Demo

- Important papers relevant to understand DALLE-2



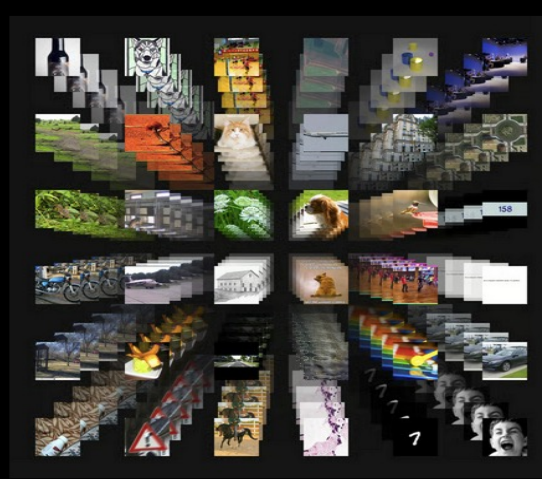
Hierarchical text-conditional image generation with CLIP latents
Apr 13, 2022



DALL-E: Creating images from text
Jan 5, 2021



DALL-E 2 pre-training mitigations
Jun 28, 2022



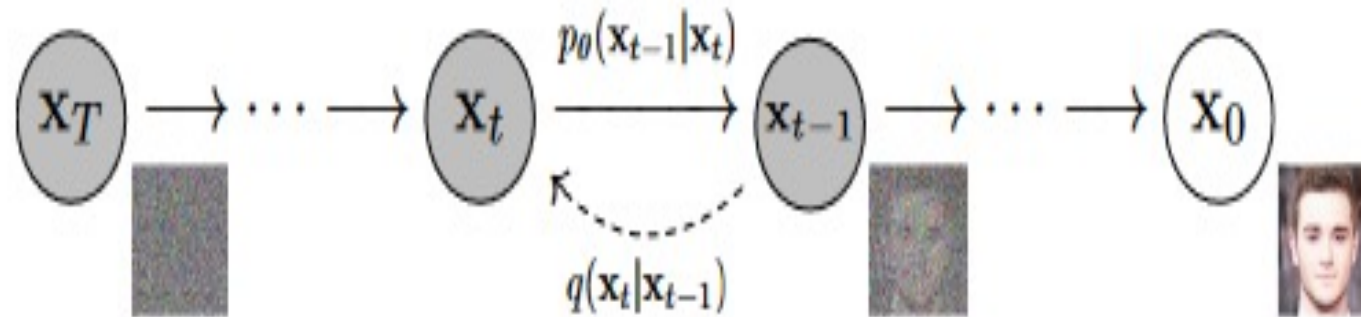
CLIP: Connecting text and images
Jan 5, 2021

- Let's get an overview of DALLE-3 and see it in action

<https://openai.com/dall-e-3>

Diffusion models (DM)

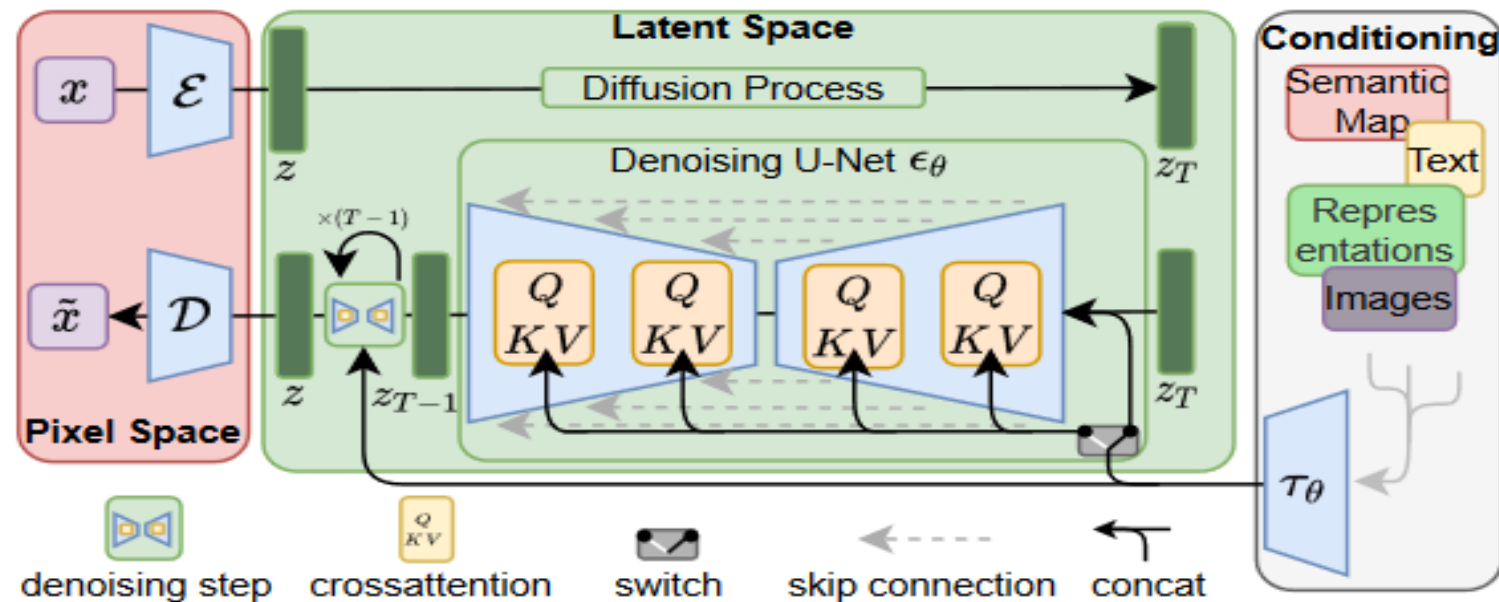
- Generative models that can generate diverse high-resolution images given a text prompt.
- Inspired by thermodynamics
- Diffusion models learn to generate data by reversing a gradual noising process
- DM learns to navigate along the parameterized Markov chain, gradually removing the noise over a series of timesteps to reverse the process of generating image from random noise.
- **Key idea:** Learning to generate images by iterative denoising



Stable Diffusion

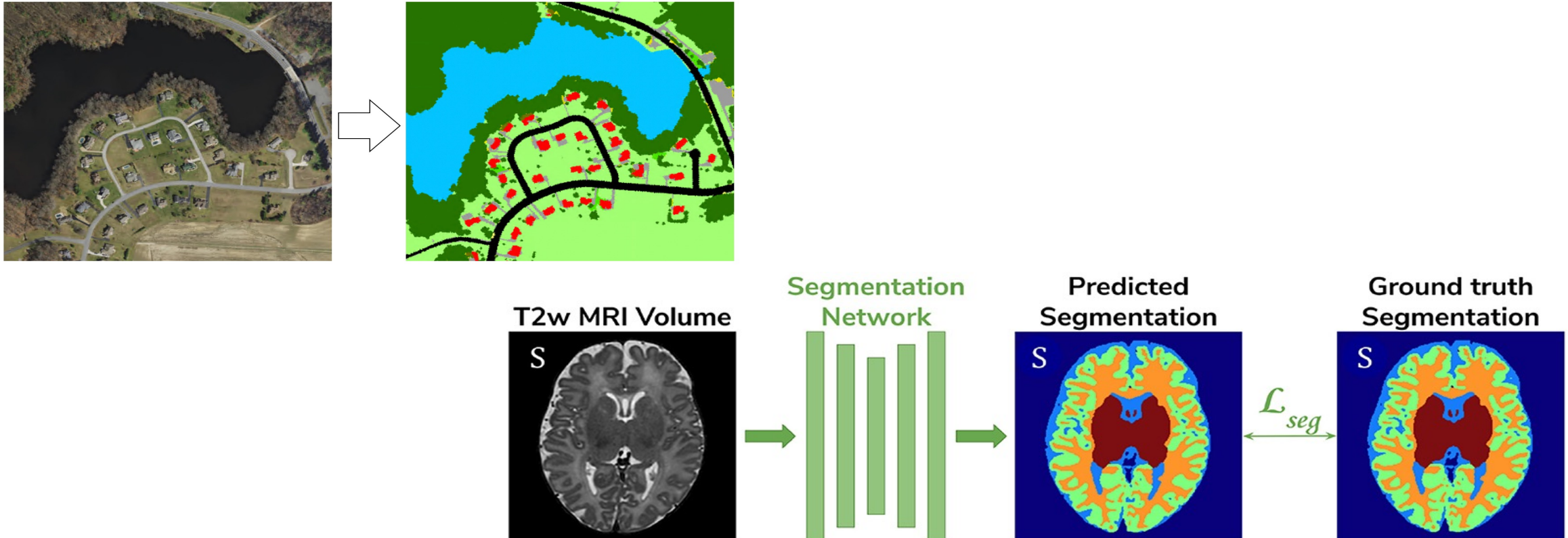
Many components integrated together to generate high-resolution images

- Forward/reverse diffusion -> method of learning to generate new stuff
- Image-text representations -> method to link images and text (e.g., CLIP method)
- Autoencoder -> method to compress images (important to speed-up process)
- U-Net + Attention -> methods to add in good inductive biases (to generate novel images)

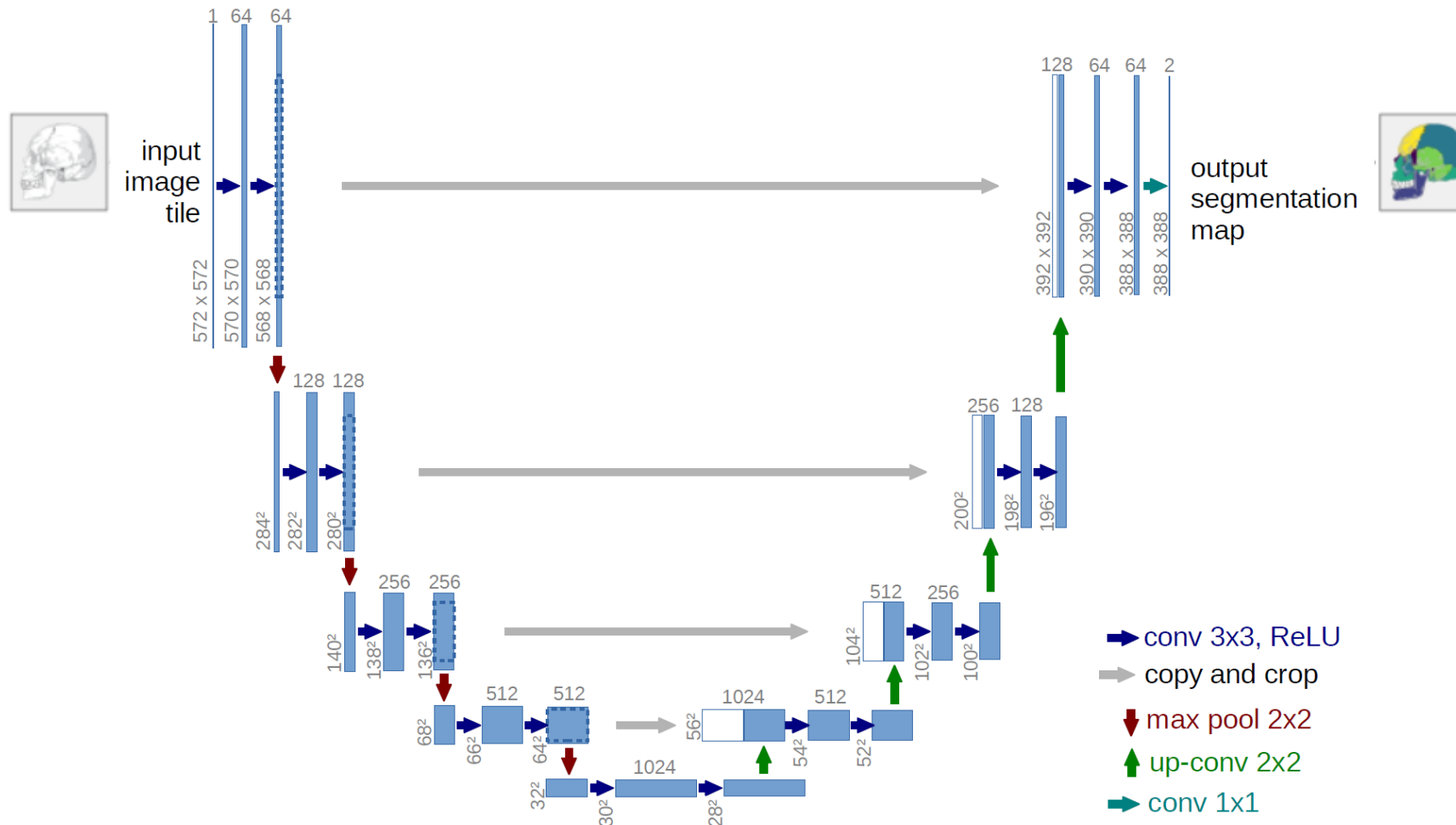


U-Net for image segmentation

- U-net learns segmentation in an end-to-end setting
- Proven to be very powerful segmentation tool in scenarios with limited annotated data
- Doesn't contain any fully connected layers



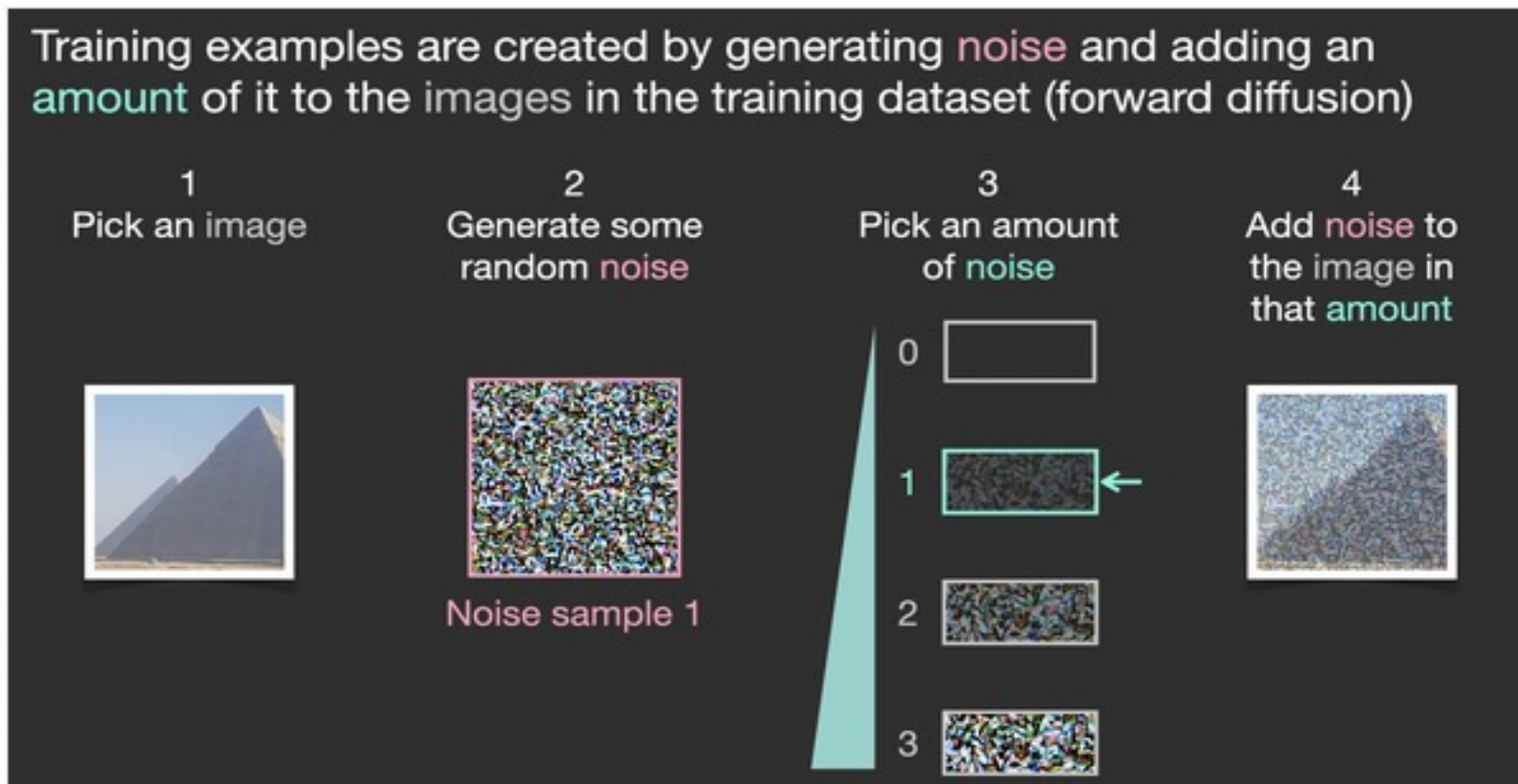
U-net Architecture



How diffusion works

Acknowledgement: Jay Alammam <https://jalammar.github.io/illustrated-stable-diffusion/>

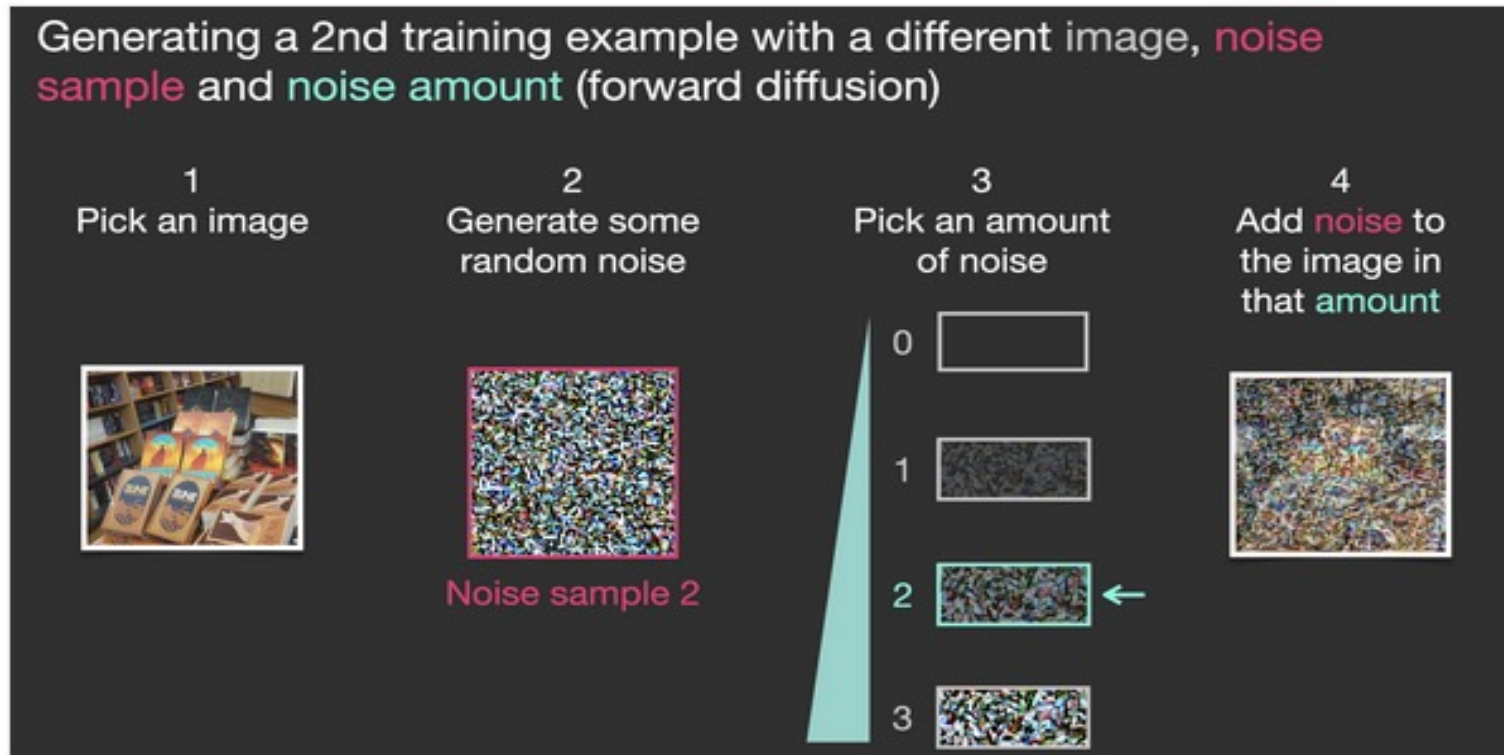
- Say we have an image, we generate some noise, and add it to the image



How diffusion works

Acknowledgement: Jay Alammam <https://jalammar.github.io/illustrated-stable-diffusion/>

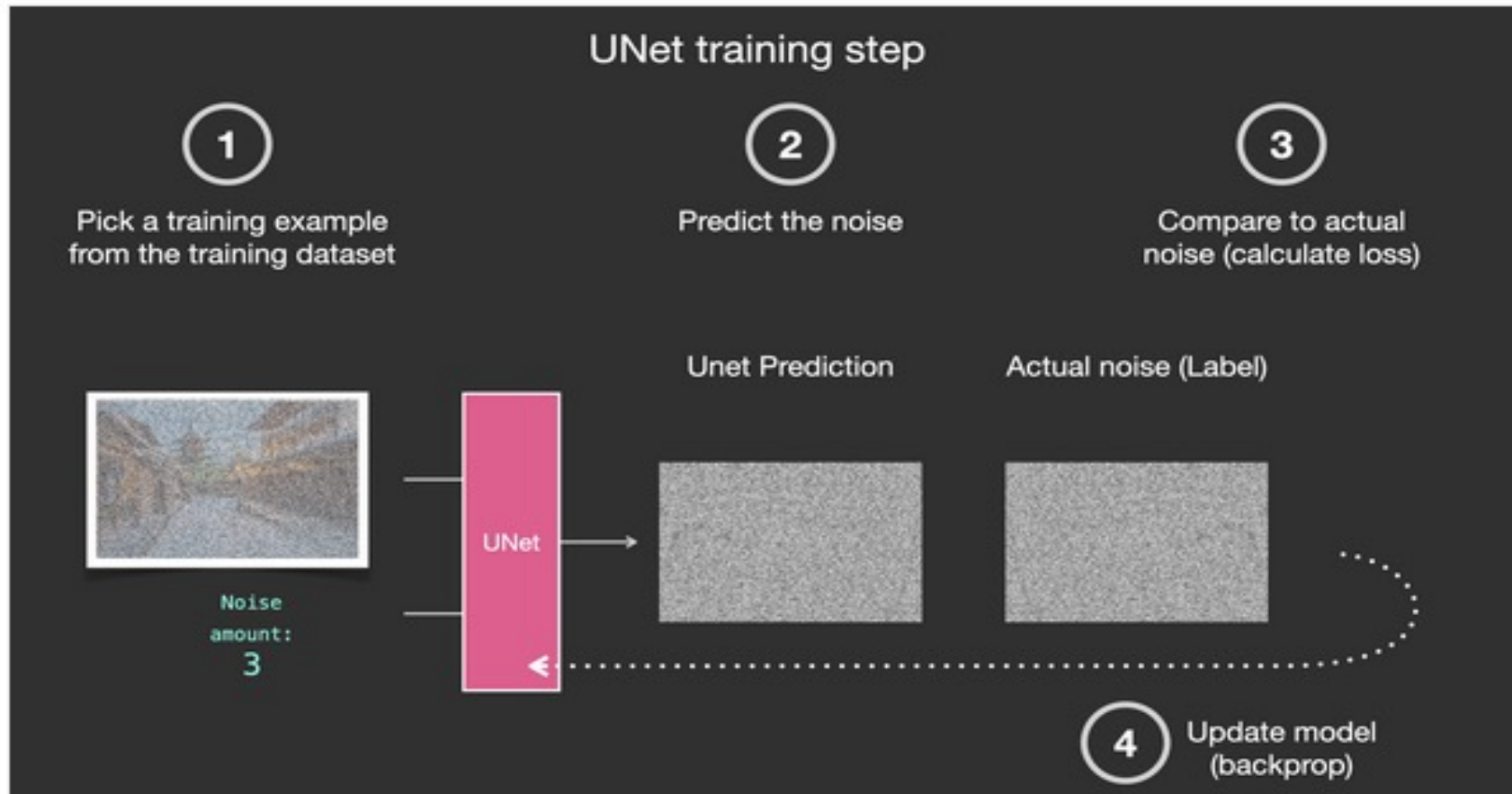
- Create lots of training examples like the same



How diffusion works

Acknowledgement: Jay Alammam <https://jalammar.github.io/illustrated-stable-diffusion/>

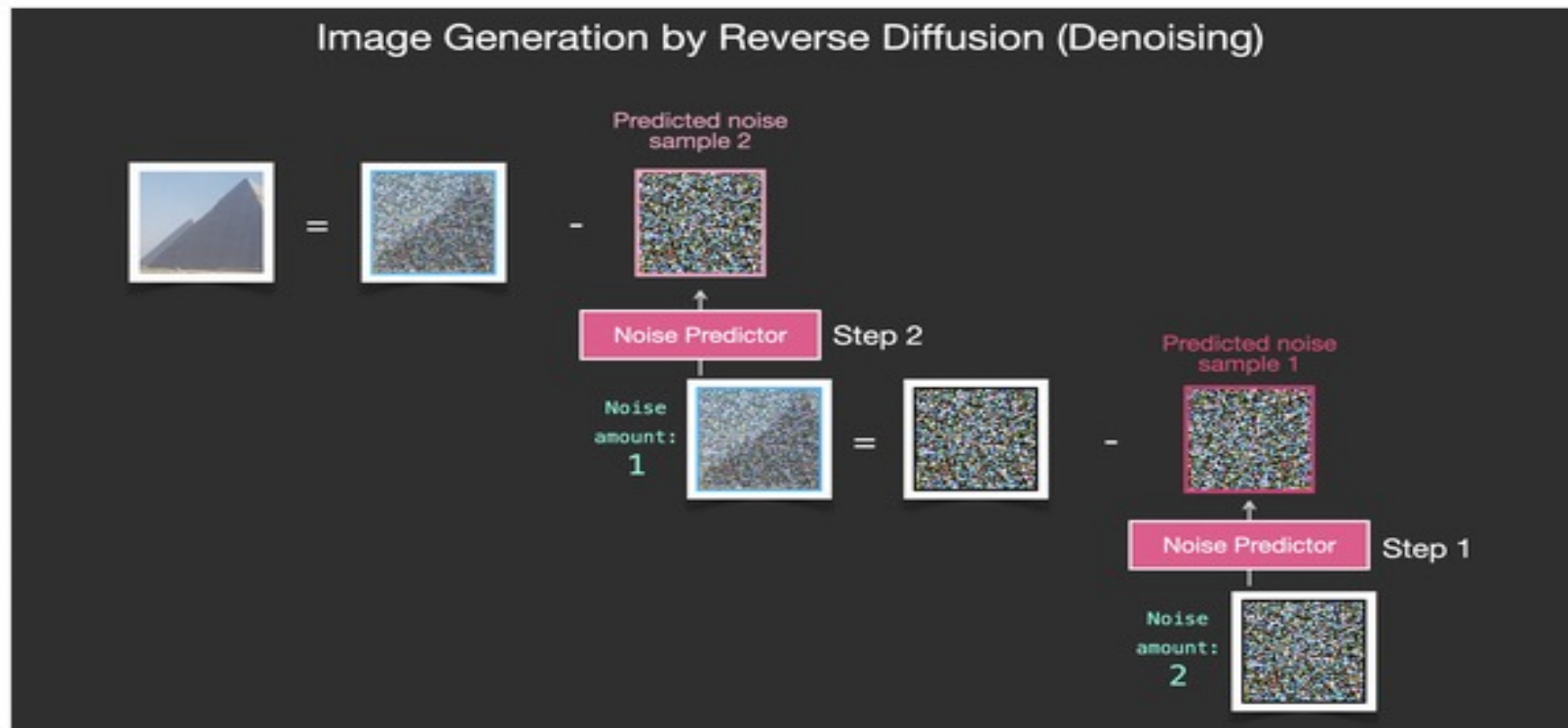
- With dataset, train the noise predictor



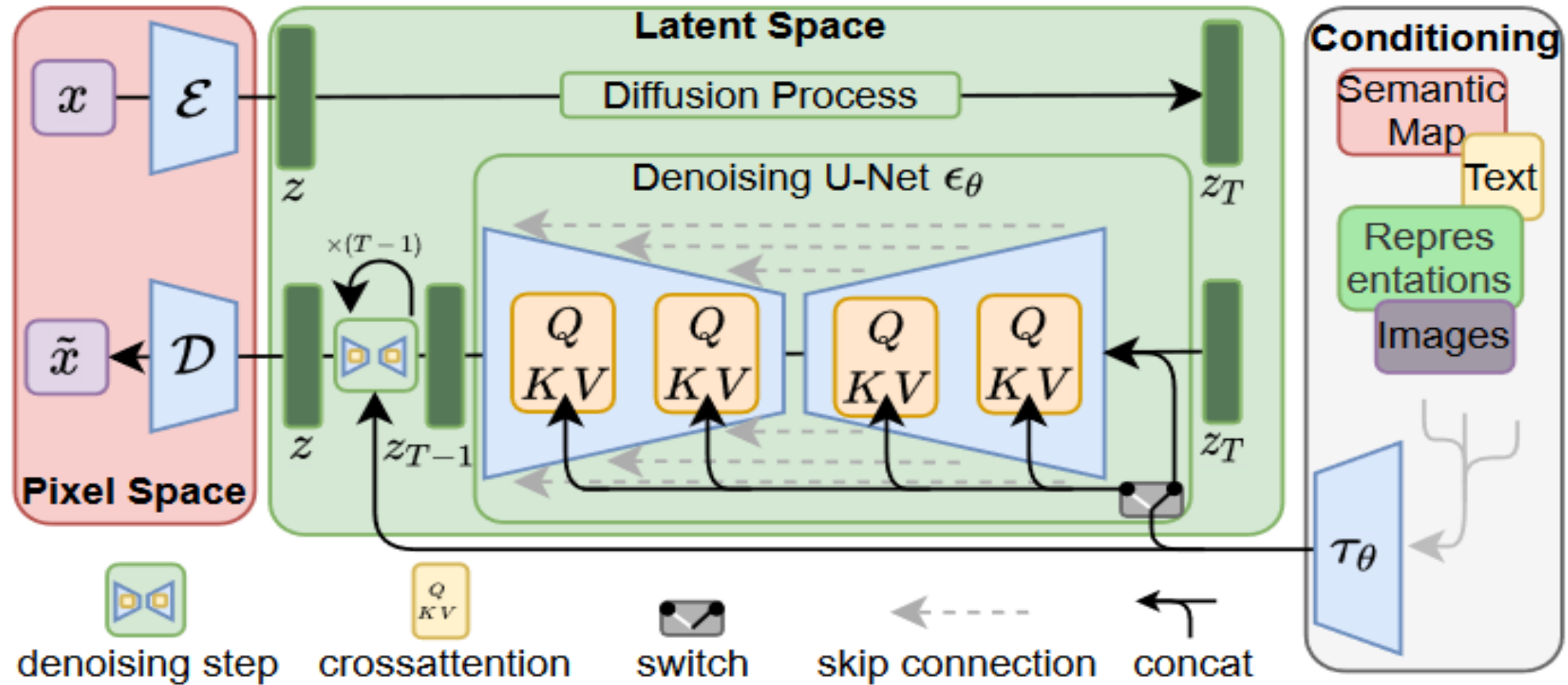
How diffusion works

Acknowledgement: Jay Alammam <https://jalammar.github.io/illustrated-stable-diffusion/>

- The trained noise predictor can take a noisy image, and with number of denoising steps, is able to predict slice of noise
- The predicted slice of noise is subtracted to get an image that is closer to images model was originally trained on (distribution of pixels)



Latent Diffusion Model/Stable Diffusion



Timeline of images generated by artificial intelligence

These people don't exist. All images were generated by artificial intelligence.

Our World
in Data

2014



Goodfellow et al. (2014) – Generative Adversarial Networks

2015



Radford, Metz, and Chintala (2015) – Unsupervised Representation Learning with Deep Convolutional GANs

2016



Liu and Tuzel (2016) – Coupled GANs

2017



Karras et al. (2017) – Progressive Growing of GANs for Improved Quality, Stability, and Variation

2018



Karras, Laine, and Aila (2018) – A Style-Based Generator Architecture for Generative Adversarial Networks

2019



Karras et al. (2019) – Analyzing and Improving the Image Quality of StyleGAN

2020



Ho, Jain, & Abbeel (2020) – Denoising Diffusion Probabilistic Models

2021

Image generated with the prompt: "a couple of people are sitting on a wood bench"



Ramesh et al. (2021) – Zero-Shot Text-to-Image Generation (OpenAI's DALL-E 1)

2022

Image generated with the prompt: "A Pomeranian is sitting on the King's throne wearing a crown. Two tiger soldiers are standing next to the throne."



Saharia et al. (2022) – Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding (Google's Imagen)

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Economic potential of Generative AI



Tweet: from Sam Altman, CEO of OpenAI



McKinsey & Company (2023):

“Generative AI is poised to unleash the next wave of productivity. ...”

Goldman Sachs (2023): “we estimate that one-fourth of current work tasks could be automated by AI in the US ... with particularly high exposures in administrative (46%) and legal (44%) professionals and low exposures in physically-intensive professions such as construction (6%) and maintenance (4%).

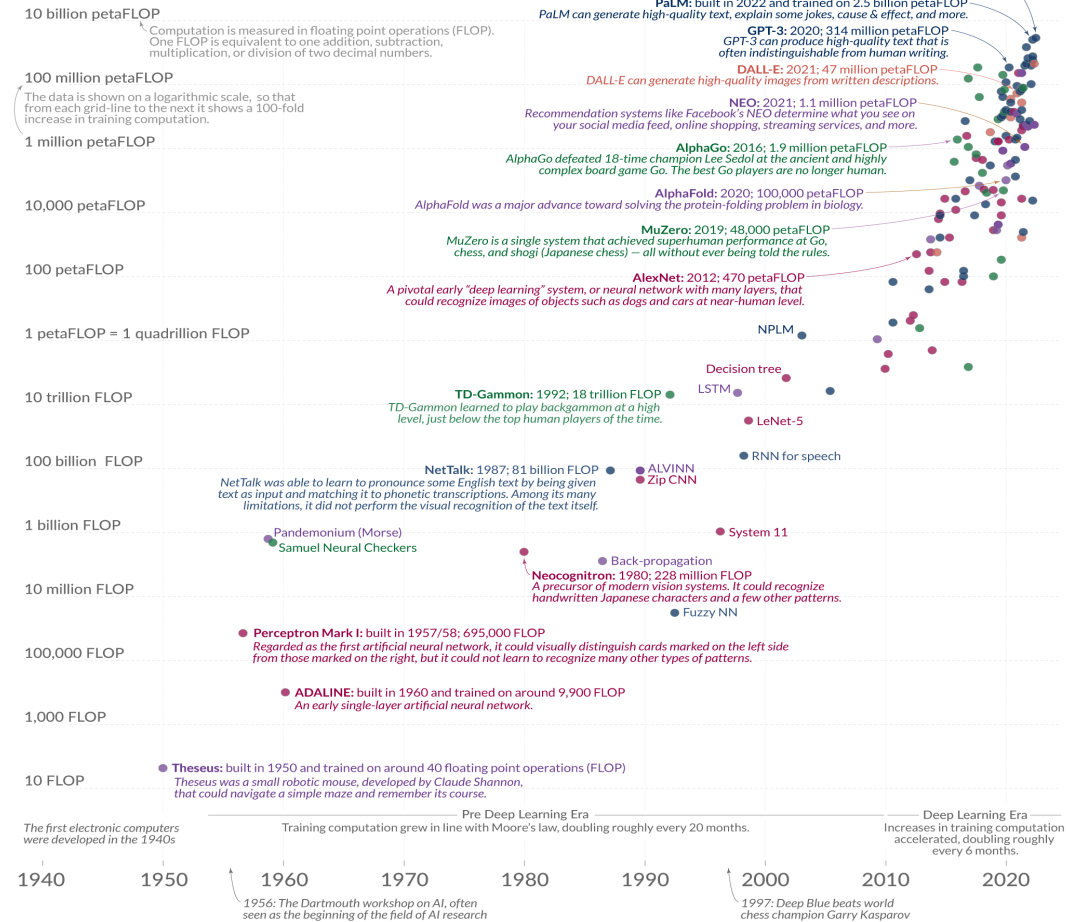
Rise of AI over the last 8 decades

The rise of artificial intelligence over the last 8 decades: As training computation has increased, AI systems have become more powerful

Our World
in Data

The color indicates the domain of the AI system: ● Vision ● Games ● Drawing ● Language ● Other

Shown on the vertical axis is the **training computation** that was used to train the AI systems.



The data on training computation is taken from Sevilla et al. (2022) – Parameter, Compute, and Data Trends in Machine Learning. It is estimated by the authors and comes with some uncertainty. The authors expect the estimates to be correct within a factor of two.

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Summary

- Generative AI has huge potential in varied applications, especially content creation.
- Generative AI can be used to:
 - formulate first drafts of promotional material
 - draft procedural correspondence
 - generate itineraries for trips
 - generate synthetic data
 - code generation (code completion, bug fixing, code style checking, code refactoring)
 - content creation for courses
 - creating designs for fashion designers
 - generating explanations for loan denials
 - multilingual customer support
 - generating automated email replies for customer support
 - Job description generation
 - draft content creation for writers
- Healthy use of generative AI will likely improve productivity across various industries.

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[10] Jay Alammar. The Illustrated Stable Diffusion

<https://jalammar.github.io/illustrated-stable-diffusion/>

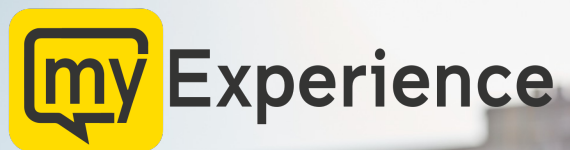
A large, abstract graphic on the left side of the slide. It consists of numerous thin, yellow, concentric lines that form a series of overlapping, irregular loops, resembling a stylized fingerprint or a topographical map. The lines are more densely packed in some areas, creating a sense of depth and movement.

Questions?



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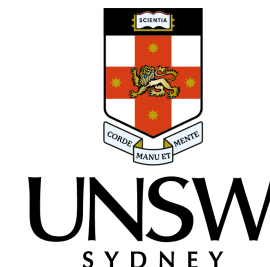
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